

Question	Answer	Marks
2(a)	(thermal) energy per unit mass (to change temperature)	B1
	(thermal) energy per unit change in temperature	B1
2(b)(i)	<i>Any three bulleted points from:</i> - the blocks end up in thermal equilibrium - heat capacity of Y is larger than heat capacity of X - no heat loss to the surroundings <i>Up to 2 points from these six:</i> - initial temperature of X = 85 °C - initial temperature of Y = 25 °C - the temperature change of X = 45 °C - the temperature change of Y = 15 °C - the temperature change in X is three times that in Y - final temperature of both = 40 °C	B3
2(b)(ii)	$\Delta\theta = 45\text{ °C}$ for X and 15 °C for Y	C1
	$mc \times 45 = 1.3 \times m \times 901 \times 15$	C1
	$c = 390\text{ J kg K}^{-1}$	A1

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